
Jelani Stewart

AI|ML|CV Engineer

New York, United States

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SUMMARY

Machine Learning/Computer Vision engineer skilled with 8 years of experience creating ML/CV models. ML/DL model architecture design from scratch and model train&test with the best performance. Implementation of the models on GCP or AWS cloud platforms and edge devices with optimization theory. Experience in computer vision algorithms for image pre-processing and enhancement with transformers and filters. Camera calibration and fusion sensor experience of RGB camera, IR sensor, and LIDAR for object detection & tracking with one-stage and two-stage detection models. Specialized in feature representation and embedding for images and input parameters.

TECHNICAL SKILLS

- Python, Java, SpringBoot, Javascript, Node.js, R,
- TensorFlow, PyTorch, Keras, Scikit-learn
- GPT-4, BERT, LangChain
- Tesseract, AWS Rekognition, OpenCV
- AWS, Docker, Kubernetes, GCP, SQL/NoSQL Databases
- Data Science, BigData
- Camera Sensors, 3D Sensors, GPU/CUDA, Nvidia Jetson

EXPERIENCE

Advance AI - *Machine Learning Engineer*

Jul 2022 - Dec 2024

Counting and registering particular objects and its different body parts flowing on a belt.

- Model selection and MLOps to production
- Training and mentoring for ML/CV algorithms and solutions

Drove an OCR project for multi-lingual environment and documents.

- Title/Caption recognition for designs, and translate it into digital format
- Layout recognition for lines' OCR
- Evaluation benchmark on different OCR algorithms and libraries
- Semi-automatic character annotation tool for handwritten characters

SweetSpot - *DL Computer Vision Engineer*

Apr 2020 - Jul 2022

Deep learning, computer vision solutions, and parallelization on CUDA

- Engineering of key point extraction and CNN-based feature representation for video compression
- Optical flow-based computer vision algorithm for keypoint movement
- CNN backbone architecture design and training for key feature extraction
- Implementation of the optical flow based algorithm and deep model on Nvidia Jetson nano device with opencv and pytorch
- Parallelizing pixel-frame quantization algorithm in CUDA library

Trax - *Machine Learning Engineer*

Aug 2018 - Apr 2020

Time series forecasting and deep language model

- Natural language token and token dictionary-based word embedding layer for feature representation
- LSTM-based bidirectional sequence-sequence model design and training to convert lip image symbols to the word sequence
- Statistical analysis for natural language specifications and prediction performance

InteractAI - *Computer Vision Algorithm Engineer*

May 2017 - Apr 2018

Computer vision and machine learning solutions for object detection and tracking Target tracking algorithm design and implementation for autonomous drone

- Data engineering and statistical analysis for improvement of siamese based tracking model
- Image enhancement and filter engineering for fusion sensor - RGB and IR camera
- Preventing track ID missing or confusing for similar targets from drone sensors with the suggestion of discriminative loss function

Person tracking and furniture detection for indoor security system

- Camera calibration for the sensor placed on top of the room
- Computer vision transformers and mathematical analysis for multi-sensor combination and object projection from image to the floor plan
- Furniture and object detection using YOLO architecture
- Person tracking with optical flow and deep model
- Body part tracking and gesture recognition with CNN model

Palo IT Singapore - *Python Developer(ML/DL)*

Aug 2017 - Apr 2018

- Implementing the papers for image retrieval service using CNN back-bone feature extraction
- Feature matching using distance metrics in feature space and PostgreSQL index configuration
- Word embedding layer training for retrieval from caption

EDUCATION

University of Pennsylvania - *Bachelor's in Artificial Intelligence*

April 2013 - March 2017